Lec-04

1. Executive Summary

- **Desktop & Local Shift:** The course transitions from web interfaces to local terminal environments using **Claude Code** for improved control, accessibility, and local file processing.
- **Instruction Files (cloud.md):** Acts as the core system prompt/guideline file for a project, establishing agent behavior and maintaining precedence over other configuration files.
- **Modular Rule Configuration:** Projects are organized using rule directories (.claude/rules) for folder-specific instructions and alternative standards like agents.md.
- **Execution Fine-Tuning:** Emphasizes auditing prompts, configuring model effort levels (Medium to Max), and maintaining structured, simple project-based workflows.

2. Core Concepts & Definitions

Term / Concept	Definition / Explanation
Claude Code	A local terminal-based agent tool serving as a collaborative partner for development tasks and exam preparation.
cloud.md	The primary instruction/system prompt file defining project-level guidelines and taking precedence in behavior conflicts.
.claude/rules	A directory structure allowing developers to define folder-specific or project-specific rules automatically followed by Claude.
agents.md	An alternative configuration file designed as a standard across multiple agent platforms, including Claude.
doc prompt audit	A diagnostic command used to review, audit, and improve prompts and instruction files.
Auto Memory	A built-in feature that automatically captures and saves project-specific memory for future context recall.
Effort Levels	Configurable execution parameters (Medium, High, Extra, Max) that control the amount of reasoning applied to a task.

3. Structured Body Hierarchy

- **Local Environment Setup & Interface Navigation**
  - **Desktop Transition:** Shifting focus from web tools to local terminal execution (claude command) for deeper machine access and better resource management.
  - **Interface Modes:** Merged workspace/chat interfaces supporting manual vs. automatic control modes, theme selection, and account binding.
- **Project Structure & Instruction Files**
  - **Project-Based Workflows:** Operating within dedicated project folders rather than random terminal directories.
  - **Instruction Hierarchy (cloud.md):** Contains general project-level directives; serves as the primary system prompt and overrides conflicting configurations.
  - **Granular Organization:** Breaking instructions down using separate guideline files, .claude/rules for directory-specific constraints, and cross-agent standards like agents.md.
- **Optimization & Best Practices**
  - **Prompt Auditing:** Regularly running doc prompt audit to refine instructions and boost effectiveness.
  - **Incremental Scaling:** Starting with simple guidelines in cloud.md and gradually expanding complexity.
  - **Model Tuning:** Adjusting effort levels and experimenting with model capabilities to optimize performance for complex tasks.

4. Review & Active Recall (Q&A)

- **Q: What is the purpose of the cloud.md file in a project?**
  - **A:** It acts as the primary system prompt and guideline file for Claude Code, shaping agent behavior within the project and taking precedence over conflicting instructions.
- **Q: Why is a project-based workflow recommended over running Claude randomly in the terminal?**
  - **A:** It ensures proper context isolation, lets the agent utilize .claude/rules and auto-memory effectively, and avoids cluttering global directory environments.
- **Q: How does the .claude/rules directory enhance project configuration?**
  - **A:** It allows developers to specify folder-specific or targeted rules that Claude automatically adheres to within designated subpaths.
- **Q: What function does the doc prompt audit command serve?**
  - **A:** It is used to review, evaluate, and iteratively improve prompts and instruction files like cloud.md.